

WMGCAMB

February 25, 2026

1 The flat special WMG Model

The Friedmann equation in the flat special WMG (eqn. (2.33) in [1]) is

$$H^2 = \frac{\mu}{3}\rho_M + 2\left(M^2 H_0^2 - \frac{\mathcal{A}_1 m_4^2}{6}\right) + 2MH_0\sqrt{\frac{\mu}{3}\rho_M + NH_0^2}, \quad (1)$$

where the subscript ‘M’ denotes ‘Matter’ (including pressureless matter, radiation, neutrinos, etc.), with

$$M = \pm \frac{\mu}{\kappa H_0}, \quad N = \left(\frac{\mu^2}{\kappa^2 H_0^2} - \frac{\mathcal{A}_1 m_4^2}{6H_0^2}\right)^2 \left(\frac{\mu}{\kappa H_0}\right)^{-2} - \frac{\Lambda_{\text{eff}}}{6H_0^2}, \quad \mathcal{C} = 0.$$

After taking the time derivative of (1), we get

$$\dot{H} = -\frac{\mu}{2}(\rho_M + P_M) \left(1 + \frac{M}{\sqrt{\frac{\mu}{3H_0^2}\rho_M + N}}\right), \quad (2)$$

where $\rho_M = -3H(\rho_M + P_M)$ has been used.

2 Modifications in CAMB

When performing the perturbation calculations, CAMB determines the Hubble parameter H and its time derivative $\dot{H}$ using the relations:

$$H^2 = \frac{\mu}{3}\rho_{\text{total}}, \quad (3)$$

$$\dot{H} = -\frac{\mu}{2}(\rho_{\text{total}} + P_{\text{total}}). \quad (4)$$

To implement the WMG model in this framework, it is useful to derive the “effective” ρ_{total} and P_{total} that satisfy these equations. Combining (2) and (4), we obtain the expression for the effective total pressure:

$$P_{\text{total}} = (\rho_M + P_M) \left(1 + \frac{M}{\sqrt{\Omega_M \frac{H^2}{H_0^2} + N}}\right) - \rho_{\text{total}}, \quad (5)$$

or equally,

$$P_{\text{total}} = P_M + (\rho_{\text{total}} - \rho_M) \left(\frac{M(\rho_M + P_M)}{\left(\sqrt{\Omega_M \frac{H^2}{H_0^2} + N}\right)(\rho_{\text{total}} - \rho_M)} - 1\right). \quad (6)$$

Specifically, the modification of ρ_{total} can be found in the Fortran file `DarkEnergyInterface.f90`’s subroutine ‘BackgroundDensityAndPressure’, where the dark energy density has been modified, contributing to the total energy density. The modification of P_{total} can be found in the Fortran file `Equation.f90`’s subroutines ‘dervis’ and ‘dervisv’ when the `gpres` is calculated.

Note:

There are some useful equations in CAMB :

$$\text{grhocrit} \equiv \frac{\mu \rho_{\text{crit}0}}{c^2} = \frac{3H_0^2}{(c/1000)^2} (\mu = 1, H_0 \text{ is in km/s/Mpc}), \quad (7)$$

$$\text{grhov} = \text{grhocrit} * (1 - \Omega_{M0}), \quad (8)$$

$$\text{grho_node_t} = \text{grhocrit} * \Omega_M(t) \frac{H^2}{H_0^2} * a^4, \quad (9)$$

$$\text{grho_matter} \equiv a^2 \mu \rho_M = \text{grhocrit} * \Omega_M(t) \frac{H^2}{H_0^2} * a^2, \text{ (the speed of light } c \text{ in } \rho_M \text{ has been supplemented)} \quad (10)$$

$$\text{gpres_node} \equiv a^2 \mu P_M = \text{grhocrit} * \Omega_M(t) \frac{H^2}{H_0^2} * w_M(t) * a^2, \quad (11)$$

$$\text{dtauda} = \frac{d\tau}{da} = \sqrt{\frac{3}{\text{grhocrit} * E^2 a^4}} = \sqrt{\frac{c^2}{H_0^2 * E^2 a^4}} = \frac{c}{a^2 H}, \quad (12)$$

$$E^2 = \frac{H^2}{H_0^2} = \left(\frac{c/1000}{a^2 H_0 \text{dtauda}} \right)^2, \quad (13)$$

$$\Omega_M(t) \frac{H^2}{H_0^2} = \frac{\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} + \dots (\text{matter part}). \quad (14)$$

In CAMB, the output `grhov_t` of the subroutine `BackgroundDensityAndPressure` is $\text{grhocrit} * a^2 * \Omega_{\text{de-eff}}(t) \frac{H^2}{H_0^2}$. From Eq.1,

$$\frac{H^2}{H_0^2} = \frac{\mu}{3H_0^2} \rho_M + 2 \left(M^2 - \frac{\mathcal{A}_1 m_4^2}{6H_0^2} \right) + 2M \sqrt{\frac{\mu}{3H_0^2} \rho_M + N}, \quad (15)$$

with the equation evaluating at today

$$1 = \frac{\mu}{3H_0^2} \rho_{M0} + 2 \left(M^2 - \frac{\mathcal{A}_1 m_4^2}{6H_0^2} \right) + 2M \sqrt{\frac{\mu}{3H_0^2} \rho_{M0} + N}, \quad (16)$$

equivalently

$$2 \left(M^2 - \frac{\mathcal{A}_1 m_4^2}{6H_0^2} \right) = 1 - \frac{\mu}{3H_0^2} \rho_{M0} - 2M \sqrt{\frac{\mu}{3H_0^2} \rho_{M0} + N}, \quad (17)$$

we can get the effective

$$\Omega_{\text{de-eff}}(t) \frac{H^2}{H_0^2} = 2 \left(M^2 - \frac{\mathcal{A}_1 m_4^2}{6H_0^2} \right) + 2M \sqrt{\frac{\mu}{3H_0^2} \rho_M + N} = 1 - \frac{\mu}{3H_0^2} \rho_{M0} - 2M \sqrt{\frac{\mu}{3H_0^2} \rho_{M0} + N} + 2M \sqrt{\frac{\mu}{3H_0^2} \rho_M + N} \quad (18)$$

$$= 1 - \Omega_{M0} - 2M \sqrt{\Omega_{M0} + N} + 2M \sqrt{\Omega_M(t) \frac{H^2}{H_0^2} + N}. \quad (19)$$

Combining the equations of `grhov` in line-93 is $\text{grhov} = \text{grhocrit} * (1 - \Omega_{M0})$, $\text{grho_node_t} = \text{grhocrit} * \Omega_M(t) \frac{H^2}{H_0^2} * a^4$ and the equation above, we can get

$$\text{grhov_t} = \text{grhocrit} * a^2 * \Omega_{\text{de-eff}}(t) \quad (20)$$

$$= (\text{grhov} - 2 * M * \sqrt{\text{grhocrit} * \sqrt{\text{grhocrit} - \text{grhov} + \text{grhocrit} * N}}) * a^2 \quad (21)$$

$$+ 2 * M * \sqrt{\text{grhocrit} * \sqrt{\text{grho_node_t} + \text{grhocrit} * N * a^4}}, \quad (22)$$

which is the output equation actually in line-93 with the `grhocrit` replaced by `crit` (here the $\text{grho_node_t} = \text{grhocrit} * \Omega_M(t) \frac{H^2}{H_0^2} * a^4$) must be preserved, otherwise the $\Omega_M(t) \frac{H^2}{H_0^2} = \frac{\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} + \dots$ term will blows up due to the $1/a$ factor during interpolation.

Note:

The `gpres` in Eq. 6 can be written by

$$\text{gpres} \equiv \mu a^2 P_{\text{tot}} = \text{gpres_node} + \frac{Ma(\text{grho_matter} + \text{gpres_node})}{\sqrt{\frac{\text{grho_matter}}{\text{grhocrit}} + N * a^2}} - \text{grhov_t}. \quad (23)$$

References

- [1] S. Garcia-Saenz, Y. Wei and X. Zhou, *Cosmology in warped massive gravity*, [[2509.09270](#)].